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|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4(a)(i)  | DNA, combined / joined, from, two / different, organisms / sources ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 |
| 4(a)(ii) | <p>any <b>three</b> from:</p> <p>(protein made in, bacteria / yeast, so)</p> <ol style="list-style-type: none"> <li>1 can be made in, large / unlimited, quantities ;</li> <li>2 cheap to make / low cost / cost-efficient ;</li> <li>3 does not, harm / involve, animals / pigs / cows ;</li> <li>4 no risk of (named) disease, transfer / infection ;</li> </ol> <p>(sequence identical to human sequence so)</p> <ol style="list-style-type: none"> <li>5 does not cause, allergy / inflammatory response / immune response / antibody production ;</li> <li>6 acts more rapidly ;</li> <li>7 do not, develop tolerance to it / need larger doses ;</li> </ol> | 3 |
| 4(b)(i)  | cow ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 |
| 4(b)(ii) | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 obtain (normal human), gene / DNA, from (m)RNA by reverse transcriptase / using restriction enzyme ;</li> <li>2 use gene editing (to alter human gene) ;</li> <li>3 obtain (nucleotide) sequence for (normal human) insulin gene ;</li> <li>4 synthesise, new / analogue, gene / nucleotide sequence ;</li> <li>5 put, new / analogue, gene in, bacteria / yeast ;</li> <li>6 add <u>promoter</u> for, gene expression / protein synthesis ;</li> <li>7 AVP ;</li> </ol>                                                                                                                   | 3 |
| 4(c)     | <p>any <b>three</b> from:</p> <ol style="list-style-type: none"> <li>1 can, share / search / access, data(base) / information / sequences ;</li> <li>2 from anywhere / online ;</li> <li>3 can, compare / align / analyse, (multiple) sequences ;</li> <li>4 rapid / fast(er) (process) ;</li> <li>5 can, quantify / count, nucleotide differences ;</li> <li>6 closely-related species have few, differences / mutations ;</li> </ol>                                                                                                                                                                                                                            | 3 |